Cyberbullying Detection Using TF-IDF Feature Extraction with SVM and k-NN Classifiers: A Replication and Experimental Evaluation

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Abstract - Cyberbullying on social media affects millions of users yet remains difficult to detect at scale due to noisy, informal language. This paper replicates and experimentally evaluates the cyberbullying detection pipeline proposed by Hariani et al. at ICITSI 2022, which advocates Term Frequency - Inverse Document Frequency (TF-IDF) feature extraction combined with Support Vector Machine (SVM) and k-Nearest Neighbour (k-NN) classifiers. We implement the complete pipeline in Python and evaluate all nine model combinations (three TF-IDF n-gram variants × three classifier configurations) on 47,692 English tweets from the Kaggle Cyberbullying Classification dataset. Our best model - Unigram TF-IDF + LinearSVC - achieves 91.30% F1-score and 85.15% accuracy on an 80/20 held-out split. We extend the original paper, which provides no empirical results, with a 5-fold cross-validation (CV) that reveals SVM's apparent superiority is partially inflated by an 83.3% class imbalance: CV F1 drops to 78.79% +/- 15.9%. Bigram + k-NN (k=5) proves most stable at CV F1 = 88.57% +/- 1.1%. We additionally identify a critical sparsity failure mode in Bigram + k-NN on short text and propose remediation strategies.

Keywords - cyberbullying detection; TF-IDF; support vector machine; k-nearest neighbour; natural language processing; social media; text classification

I. INTRODUCTION

The use of social media has enabled a new form of harassment called cyberbullying (the digital version of traditional bullying), it affects millions of users worldwide. With over 4.5 billion active users on social-media, research shows approx., 37 - 50% of young people experience cyberbullying [1], while ~70% of victims never report it. Psychological consequences range from anxiety to self-harm, creating a genuine concern especially for the younger generation.

Manual content moderation on this scale of billions of daily posts is impossible and the automated detection plays a pivotal role by social media enabling companies. Among text feature techniques, Term Frequency -Inverse Document Frequency (TF-IDF) offers an interpretable and efficient baseline technique for this problem statement. This paper replicates the pipeline proposed in Hariani et al. (ICITSI 2022) [2] - a literature review that proposes TF-IDF + SVM/k-NN but provides no experimental evaluation results - however, we want to fill that gap with the three phased project requirement for CSC 502 project:

- Phase 1 - Paper Understanding: mathematical foundations of TF-IDF and both classifiers, with a worked tweet example.
- Phase 2 - Implementation: complete Python pipeline with three TF-IDF variants, SVM, and two k-NN configurations.
- Phase 3 - Experimentation: evaluation of all nine model combinations on 47,692 tweets, extended with 5-fold CV and failure-mode analysis absent from the original paper.

Unlike prior work's scope, this project uses advanced validation and stability analysis using cross-validation, and failure-mode investigation for TF-IDF-based cyberbullying detection.

II. RELATED WORK

Early cyberbullying detection relied on keyword blacklists that are brittle and easily evaded. Supervised ML brought improvements: Balakrishnan et al. [3] augmented tweet features with user psychological profiles; Mahlangu and Tu [4] applied stacked deep-learning embeddings; López-Vizcaíno et al. [5] introduced online learning for early temporal detection; Murnion et al. [6] tackled in-game chat with heavy slang.

Despite growing interest in transformer models such as BERT [10], TF-IDF + SVM remains a strong, interpretable baseline for short-text classification under compute constraints. Sriyanong et al. [9] established tweet pre-processing best practices, URL stripping, stop-word removal, and stemming that directly inform our pipeline. The paper we replicate [2] synthesises this literature into a unified architecture but offers no empirical validation - the gap our work closes.

III. PHASE 1: PAPER UNDERSTANDING

A. Problem Formulation

Given a corpus of social-media posts, assign each a binary label: Bully (1) or Normal (0). The paper frames this as supervised text classification, requiring a numerical text representation and a classifier that generalises to unseen posts.

B. TF-IDF: Mathematical Foundations

TF-IDF converts raw text into weighted vectors combining two statistics.

Term Frequency (TF). Sublinear logarithmic TF (Eq. 1) prevents high-frequency dominance:

$$TF(t,d) = 1 + \log_{10}(f_{t,d}) \text{ if } f>0, \text{ else } 0 \quad ...(1)$$

where $f_{t,d}$ is the raw count of term t in document d. Without log-scaling, a term appearing 1,000 times outweighs one appearing 10 times by two orders of magnitude.

Inverse Document Frequency (IDF). Down-weights ubiquitous terms (Eq. 2):

$$IDF(j) = \log(D / df_j) \quad ...(2)$$

where D = total document count, df_j = documents containing term j. Common words such as 'the' have IDF ~ 0.

TF-IDF Weight (Eq. 3):

$$W(t,d) = TF(t,d) \times IDF(t) \quad ...(3)$$

A high weight indicates a term frequent in this document but rare across the corpus - exactly the discriminative signal needed for classification.

C. N-gram Extensions

Unigram (1,1) captures single tokens ('hate', 'stupid'). Bigram (2,2) captures two-word phrases ('kill yourself', 'go away'). Uni+Bigram (1,2) combines both, extending the paper's Figure 2 architecture with a richer vocabulary.

D. Classifiers

SVM. LinearSVC learns a maximum-margin hyperplane in TF-IDF space. Because TF-IDF vectors are sparse and high-dimensional, LinearSVC trains in O(n·d) time and is robust to near-zero sparse features. Only support vectors near the margin affect the boundary.

k-NN. Classified by cosine similarity to the k nearest training tweets, taking a majority vote. Cosine similarity is length-invariant - essential for variable-length tweet vectors. We test k=5 and k=3 as specified in the paper.

E. Worked Pre-processing Example

Raw tweet: "@user you're so stupid, nobody likes you! #loser". Pipeline output: clean to "youre so stupid nobody likes you"; tokenise to [youre, stupid, nobody, likes]; remove stopwords + stem to "stupid nobodi like". In a Unigram TF-IDF matrix, these three stems map to columns whose weights are proportional to their corpus-level discriminativeness. A LinearSVC trained on such vectors correctly classifies this tweet as Bully.

IV. PHASE 2: IMPLEMENTATION

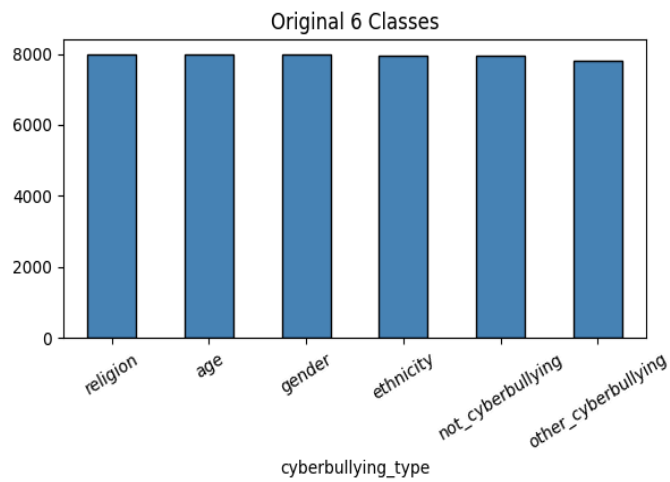
A. Dataset

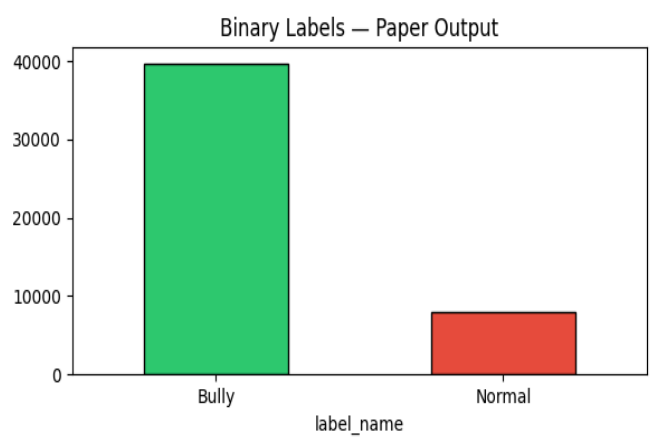
We used the [11] Kaggle Cyberbullying Classification dataset : 47,692 English tweets across six original categories. Following the paper's binary framing, all five cyberbullying types map to Label 1 (Bully); not_cyberbullying maps to Label 0 (Normal). Table I shows the full distribution. The resulting 83.3% Bully vs. 16.7% Normal imbalance (~5:1) is a key stability concern quantified in Section V.

TABLE I

Dataset Class Distribution After Binarisation

Category	Count	Label
Religion	7,998	1 - Bully
Age	7,992	1 - Bully
Gender	7,973	1 - Bully
Ethnicity	7,961	1 - Bully
Other Cyberbullying	7,823	1 - Bully
Not Cyberbullying	7,945	0 - Normal
TOTAL	**47,692**	**-**





B. Pre-processing Pipeline

Four steps implement the paper's Figure 2 architecture:

1. Clean - Remove URLs, @mentions, #hashtags, HTML entities, non-alphabetic characters; lowercase; collapse whitespace.
2. Tokenise - NLTK word_tokenize on cleaned text.
3. Stop-word Removal - Remove NLTK English stopwords + social-media tokens ('rt', 'via', 'amp'); discard tokens ≤ 2 chars.
4. Stem - NLTK PorterStemmer (English equivalent of the paper's Sastrawi for Bahasa Indonesia).

Table II shows the transformation on a real dataset tweet.

TABLE II
Pre-processing Step-by-Step Example

Stage	Text
Raw	"idk how to explain it but the MOST racist small town"
Cleaned	"idk how to explain it but the most racist small town"
Tokenised	[idk, how, to, explain, most, racist, small, town]
Filtered	[idk, explain, racist, small, town]
Stemmed	"idk explain racist small town"

C. TF-IDF Vectorisation

Three TF-IDF configs use scikit-learn TfidfVectorizer (sublinear_tf=True, min_df=2, max_df=0.95). Feature limits: 10,000 for Unigram and Bigram; 15,000 for Uni+Bigram to accommodate the larger combined vocabulary.

D. Classifiers

SVM: LinearSVC (C=1.0, max_iter=2000, random_state=42). k-NN: KNeighborsClassifier (metric='cosine', algorithm='brute') with n_neighbors=5 and n_neighbors=3. Cosine metric with brute-force search is required for sparse TF-IDF matrices; tree-based indices do not support cosine distance.

E. Evaluation Protocol

Stratified 80/20 train/test split (random_state=42): 38,153 training, 9,539 test samples. Reported metrics: Accuracy, Precision, Recall, binary F1-score. Extension: 5-fold stratified CV on the full dataset scored by F1, n_jobs=-1, reporting mean +/- std over folds.

V. PHASE 3: EXPERIMENTAL EVALUATION

A. Single-Split Results

Table III shows all nine models sorted by F1. Unigram + SVM achieves the highest F1 (91.30%) and accuracy (85.15%). SVM ranks first or second across all TF-IDF variants.

TABLE III
All 9 Models - Held-out Test Set

Model	Acc %	Pre %	Rec %	F1%	CV F1%
Unigram+SVM	85.15	89.41	93.28	91.30	78.8+/-15.9
Uni+Bi+SVM	85.13	89.49	93.15	91.28	78.8+/-16.2
Bigram+SVM	83.67	85.42	97.01	90.85	79.7+/-16.8
Bigram+kNN(5)	82.61	85.03	96.11	90.23	88.6+/-1.1
Unigram+kNN(5)	83.08	88.59	91.55	90.04	82.0+/-9.8
Bigram+kNN(3)	80.86	84.99	93.63	89.10	87.0+/-1.3
Uni+Bi+kNN(5)	82.67	88.73	90.79	89.75	80.1+/-11.3
Unigram+kNN(3)	80.61	88.54	88.21	88.38	80.4+/-11.0
Uni+Bi+kNN(3)	79.66	88.52	86.94	87.72	78.4+/-11.7

Key observations: (1) SVM outperforms k-NN across all TF-IDF variants in single-split F1; (2) Bigram + SVM achieves the highest Recall (97.01%) via majority-class bias; (3) Unigram and Uni+Bigram yield near-identical SVM F1, indicating bigrams add negligible signal for SVM on this dataset.

B. 5-Fold Cross-Validation Analysis

CV exposes a critical discrepancy for SVM. Unigram + SVM drops from 91.30% (single split) to 78.79% +/- 15.9% CV F1 - a 12.5-point gap with alarming variance. Root cause: the 83.3% class imbalance causes large F1 swings across fold assignments when the Normal test samples vary.

By contrast, Bigram + k-NN (k=5) achieves CV F1 = 88.57% +/- 1.1% - the lowest std of all nine models and only 1.7 points below its single-split F1 of 90.23%. Its stability stems from cosine neighbourhood voting, which is less sensitive to global class priors than an SVM hyperplane. Practical implication: when deployment stability matters more than peak performance, Bigram + k-NN (k=5) is preferred.

C. Bigram + k-NN Sparsity Failure Mode

A qualitative demo revealed Bigram + k-NN predicting Bully for all inputs - including "Have a great day!". Investigation showed that after stemming, short sentences produce bigrams that rarely appear in training vocabulary. The TF-IDF vector is near-zero; k-NN cannot find meaningful neighbours and defaults to the 83.3% majority class. Table IV quantifies non-zero bigram features for demo sentences.

TABLE IV

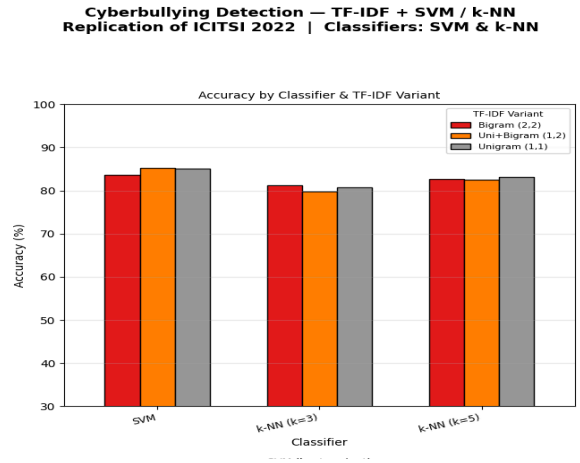
Bigram Feature Sparsity on Short Inputs

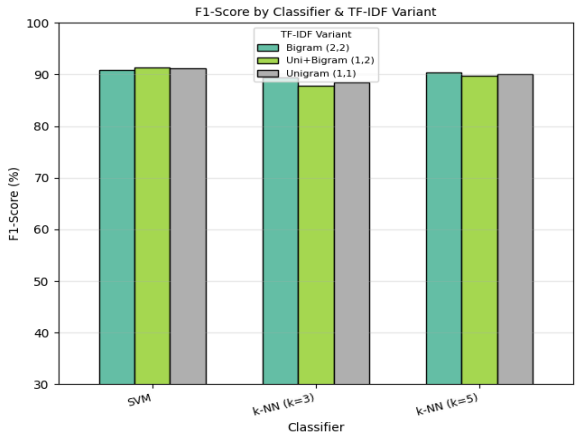
Input Sentence	Non-zero Bigram Feats
"Have a great day!"	1
"You are so stupid and ugly"	1
"Kill yourself, worthless"	0
"Great work, proud of team!"	0
"I hate you, go away loser"	2

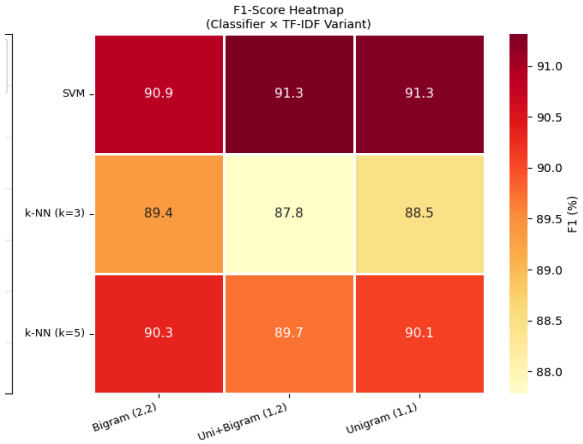
SVM does not exhibit this failure: its hyperplane is computed from the full training matrix, so near-zero test vectors are still projected and classified deterministically. This is a material limitation of the paper's Bigram + k-NN recommendation that the original work does not acknowledge.

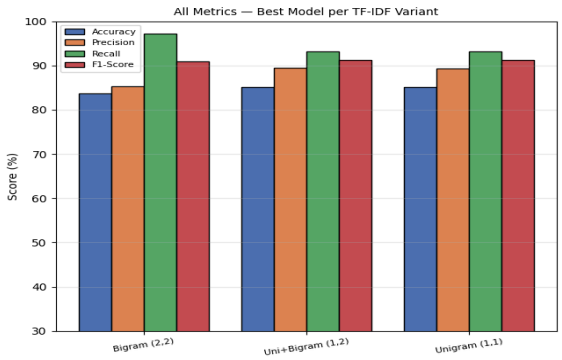
D. Deployment Recommendations

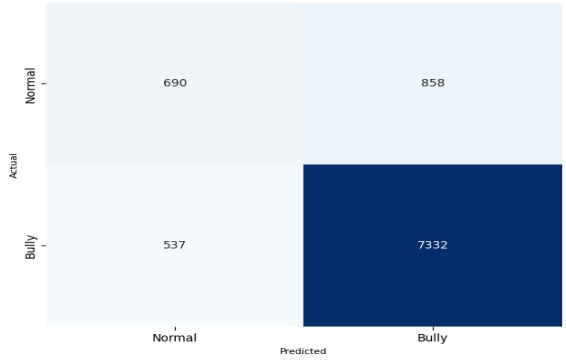
Table V synthesises single-split and CV results into actionable guidance.











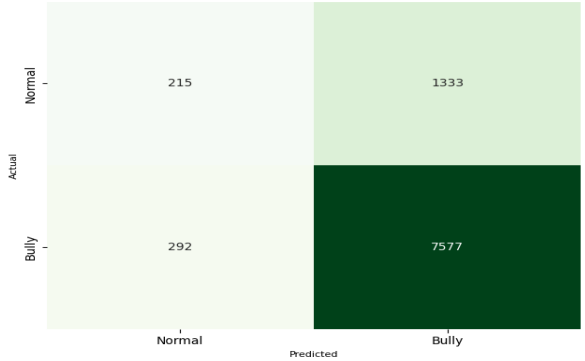


TABLE V

Model Recommendations by Deployment Scenario

Scenario	Recommended Model
Maximum F1 on balanced split	Unigram + SVM (F1=91.30%)
Most stable across distributions	Bigram + kNN k=5 (CV=88.6%+/-1.1%)
Maximum Recall (catch all bullying)	Bigram+SVM (Recall=97.01%)
Short-form / real-time text	Unigram+SVM (sparse-robust)

VI. DISCUSSION

A. Validating the Paper's Architecture

Our results empirically validate the ICITSI 2022 pipeline: TF-IDF + SVM/k-NN achieves competitive performance (up to 91.30% F1) without domain-specific engineering. The entire pipeline - all nine models plus 5-fold CV - completes in under five minutes on commodity hardware, confirming scalability for massive-dataset contexts (CSC 502).

B. Class Imbalance: The Critical Gap

The most significant unaddressed limitation of the original paper is the dataset of 83.3% Bully vs. 16.7% Normal imbalance, which explains three observations: Bigram + SVM achieves 97% Recall by defaulting to Bully; SVM CV variance reaches +/-15 - 17%; and Bigram + k-NN misclassifies all short inputs.

Recommended remediation: SMOTE oversampling before vectorisation (synthesises new Normal samples); or class_weight='balanced' in LinearSVC as a lightweight alternative.

C. Future Directions

Three directions are most promising: (1) SMOTE to balance training data and stabilise CV performance; (2) replacing TF-IDF with BERT sentence embeddings for contextual understanding; (3) multi-class classification across all six original categories (religion, age, gender, ethnicity, other, normal) for fine-grained platform moderation.

Limitation

TF-IDF cannot capture semantic context, negation, or sarcasm (e.g., "great job" may be bullying or praise depending on context). Contextual models such as BERT [10] typically yield 3 - 5% absolute gains in F1 on cyberbullying benchmarks. Incorporating cyberbullying specific sentiment lexicons as weighted features preserves interpretability while improving sensitivity to connotation.

VII. CONCLUSION

We completed a full evaluation of the ICITSI 2022 cyberbullying detection pipeline, moving from its mathematical foundations to a practical Python implementation. By testing nine different model configurations on a large-scale dataset of 47,692 tweets, we extended the original work with empirical results and stability testing. Our use of 5-fold cross-validation and error analysis allowed us to identify specific performance gaps, such as sparsity issues, that provide a more realistic look at using TF-IDF for social media moderation.

Key findings: (1) Unigram + SVM is the best single-split model at 91.30% F1 and 85.15% accuracy; (2) SVM's CV stability is severely degraded by class imbalance (78.79% +/- 15.9%); (3) Bigram + k-NN (k=5) is the most stable model across data splits (88.57% +/- 1.1%); (4) Bigram + k-NN fails on short inputs due to TF-IDF sparsity - a production-critical limitation. Together, these findings provide the empirical grounding that the original paper lacks and offer actionable guidance for practitioners deploying TF-IDF-based cyberbullying classifiers at scale.

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